

Numerical Analysis Assignment 4

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November 2024

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1 Problem 3.5a

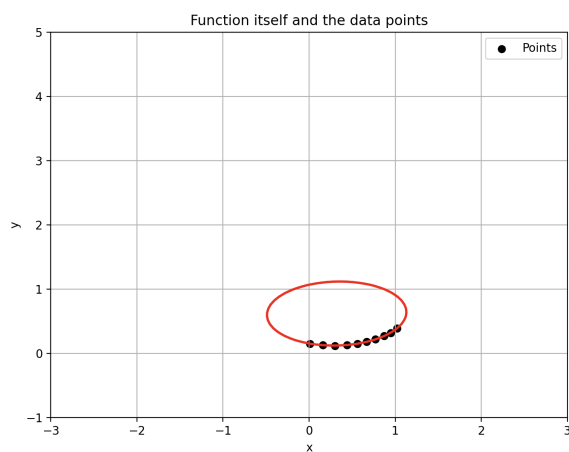


Figure 1: Problem 3.5a

```
Parameters:  
a = -2.6356254837120896  
b = 0.14364618259888695  
c = 0.55144696314036  
d = 3.222940338105897  
e = -0.4328942702644511
```

Figure 2: Parameters of the Ellipse

2 Problem 3.5b

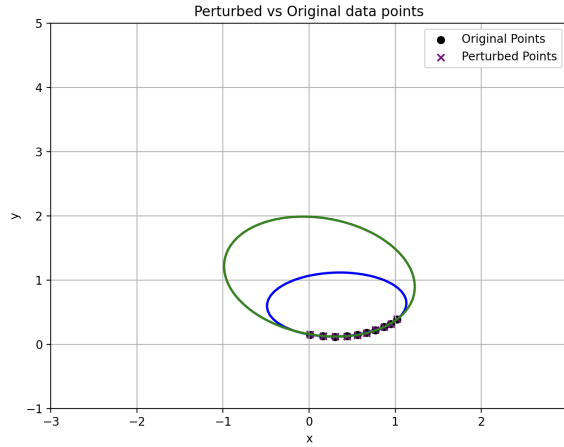


Figure 3: Problem 3.5b

```

Perturbed Parameters:
a_per = -1.4051270147476034
b_per = -0.4067920792051215
c_per = 0.6694871307461846
d_per = 3.0155544743074554
e_per = -0.4416717136116998

Difference between Original and Perturbed Parameters:
Delta a = 1.2304984689644862
Delta b = -0.5504382618040085
Delta c = 0.11804016760582459
Delta d = -0.2073858637984416
Delta e = -0.008777443347248726

```

Figure 4: Perturbed parameters and the original parameters relation

In the plot, the green plot is the perturbed orbit, and the blue one is original plot. We can also see that, after the perturbation on the matrix, the parameters have changed a lot. For this reason, the plot of the orbit became very different from the previous.

The reason for such a big change in the parameters is that the condition number of the matrix is high as the matrix is nearly rank deficient.

3 Problem 3.11

3.1 Matrix Q Formation

In my code, I used `householder_qr` function to create Reduced Echelon form and vectors of Householders. To calculate Reduced Echelon form, we apply proper Householder transformation to the each matrix column in sequence. This way, each transformation annihilates the entries below the diagonal entry inside one of the columns of the matrix. So, we can get the Reduced Echelon form if we apply the matrices in sufficient number. To get the vectors to represent these Householder transformations, the following equation is used, where $\alpha = -\text{sign}(a_k) \|a_2\|_2$:

$$\mathbf{v} = \begin{bmatrix} \mathbf{o} \\ \mathbf{a}_2 \end{bmatrix} - \alpha \mathbf{e}_k,$$

Figure 5: Householder vector

Here, $\mathbf{a}_2$ is the vector below the diagonal entry. Moreover, `make_q` takes the vectors of H as inputs and then applies Householder transformations on identity matrix in order to construct multiplications of Householder matrices. As

$$H_n \dots H_1 A = R$$

We can see that $H_n \dots H_1 Q R = R$. Then, we can infer that this means $H_n \dots H_1 Q = I$. So, multiplying each side with Q^T from right, and taking transpose of each side, we would get. $Q = (H_n \dots H_1)^T$. This way, we calculate the Q matrix.

Also, Householder vectors are stored efficiently. To elaborate, the entries below diagonal entries excluding diagonal entries stores the House-

holder vectors (except diagonal). And we create another column to save the remaining entries.

The following picture includes the orthogonal matrix Q and upper triangular matrix R . We can see that indeed the forms of the matrices are compatible.

Also, for testing, I randomly (from Standard distribution) generated the entries of A .

```
Orthogonal matrix Q:
[[-0.698277 -0.386409  0.417505  0.032485  0.43328 ]
 [ 0.15283  -0.340081 -0.569492 -0.506039  0.529708]
 [-0.05603  -0.087622 -0.485151  0.835417  0.236411]
 [ 0.629724 -0.627773  0.408618  0.200606  0.046224]
 [ 0.298932  0.577284  0.314688  0.068587  0.688223]]

Upper triangular matrix R:
[[-3.810388  0.43247  1.324559]
 [-0.        -2.601514 -1.448845]
 [ 0.         0.        -1.638404]
 [-0.         0.         0.         ]
 [-0.         -0.         0.         ]]

Is Q orthogonal? True
```

Figure 6: Results of Problem 3.11

In the following picture includes the verification of the results shown in the picture above. We can see that multiplying transpose of Q by Q results in identity matrix, which means that Q is orthogonal. Moreover, we can see that the original matrix and the matrix obtained by multiplying Q with R are same. This means that the QR decomposition is done correctly.

```

Q^T * Q:
[[ 1. -0.  0.  0.  0.]
 [-0.  1. -0. -0.  0.]
 [ 0. -0.  1. -0.  0.]
 [ 0. -0. -0.  1.  0.]
 [ 0.  0.  0.  0.  1.]]

Original matrix A:
[[ 2.6607054  0.70326526 -1.04910341]
 [-0.58234079  0.95081875  1.62821476]
 [ 0.21349478  0.20371769  0.84760836]
 [-2.39949434  1.90549774  1.07417303]
 [-1.13904702 -1.37253191 -0.95602639]]

Reconstructed matrix Q * R:
[[ 2.660705  0.703265 -1.049103]
 [-0.582341  0.950819  1.628215]
 [ 0.213495  0.203718  0.847608]
 [-2.399494  1.905498  1.074173]
 [-1.139047 -1.372532 -0.956026]]

```

Figure 7: Verification of Problem 3.11

4 Problem 4.1

```

Eigenvalues of A: [ 2.00000000e+00 -2.08166817e-17]
Eigenvectors of A:
[[ 9.999995e-01 -9.999995e-01]
 [ 9.999995e-04  9.999995e-04]]

Condition number of eigenvector matrix of A: 1000.0000000000001
Absolute condition numbers for each eigenvalue of A: [500.0005, 500.00050000000005]

Eigenvalues of B: [1. 1.]

Comparison of Eigenvalues of A and B:
Eigenvalue of A: 2.0 , Eigenvalue of B: 1.0
Eigenvalue of A: -2.0816681711721685e-17 , Eigenvalue of B: 1.0

```

Figure 8: Result of Problem 4.1

The change in the Eigenvalues of the matrix is reasonable as the condition number is very high. Previously the eigenvalues were 2.0 and -2.0 but

then it became 1.0 and 1.0. When the rank of a matrix is close to being deficient (almost linearly dependent rows or columns), its eigenvalues can be unstable and highly sensitive to perturbations. Also, we could have guessed that the matrix has high condition number as the values in the matrix are very different from each other.